

SapBERT Four-Level Comparison

Sections 3.4 and 3.5 | Full 768-D versus PCA-64

Encoder: SapBERT. Frozen, normalized 768-dimensional clinical-concept embeddings. PCA reduces the same embeddings to 64 dimensions. Each classifier is evaluated on both text-only and text-plus-patient inputs, producing all 12 comparisons.

Dataset: 10,000 supplied cardiac records. Fixed grouped split: 8,001 development and 1,999 test rows; labels 0-3 (Emergency, Urgent, Standard, Non-urgent). Every comparison uses the same rows. PCA and patient-feature preprocessing are fitted on development rows only. Classifier settings are fixed across representations.

Class totals: L0 Emergency: 2,040; L1 Urgent: 2,714; L2 Standard: 3,000; L3 Non-urgent: 2,246.

Text-only comparison

Text size	Classifier	Accuracy	Precision*	Recall*	F1*
768-D	Logistic Regression	44.17%	44.89%	44.87%	43.95%
768-D	Hist Gradient Boosting	41.52%	43.72%	40.95%	41.70%
768-D	Random Forest	42.92%	44.82%	42.93%	43.49%
PCA-64	Logistic Regression	43.42%	44.74%	44.20%	43.26%
PCA-64	Hist Gradient Boosting	42.57%	45.34%	41.85%	42.80%
PCA-64	Random Forest	40.62%	41.88%	40.93%	41.08%

Text plus patient features

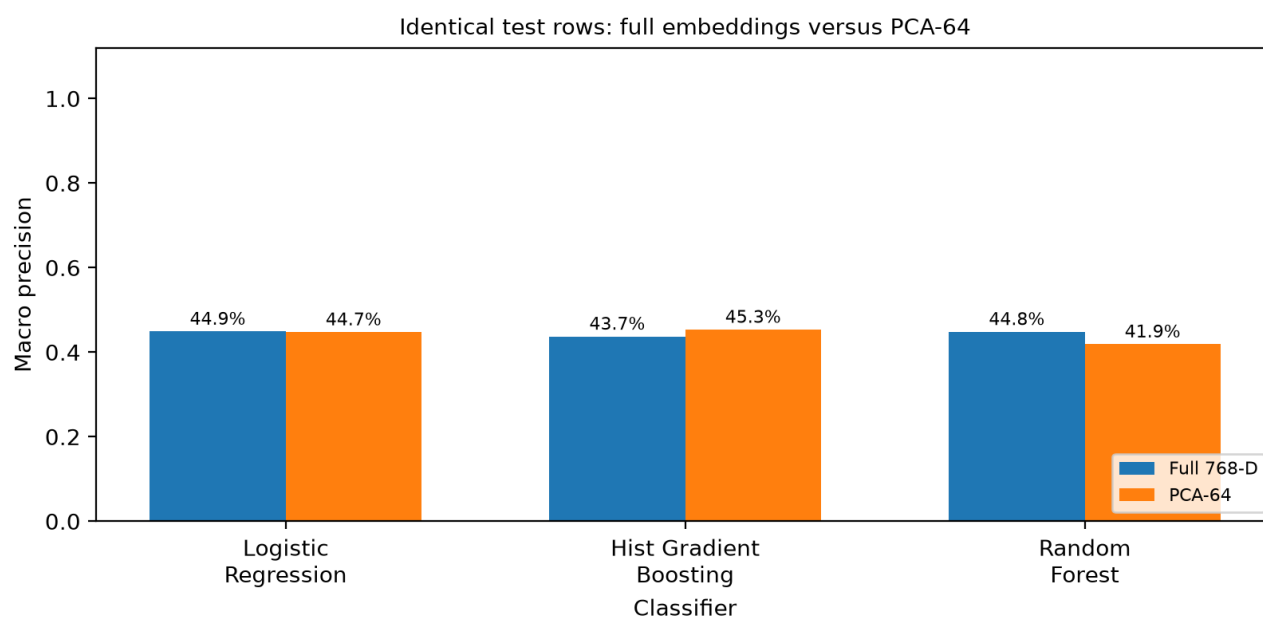
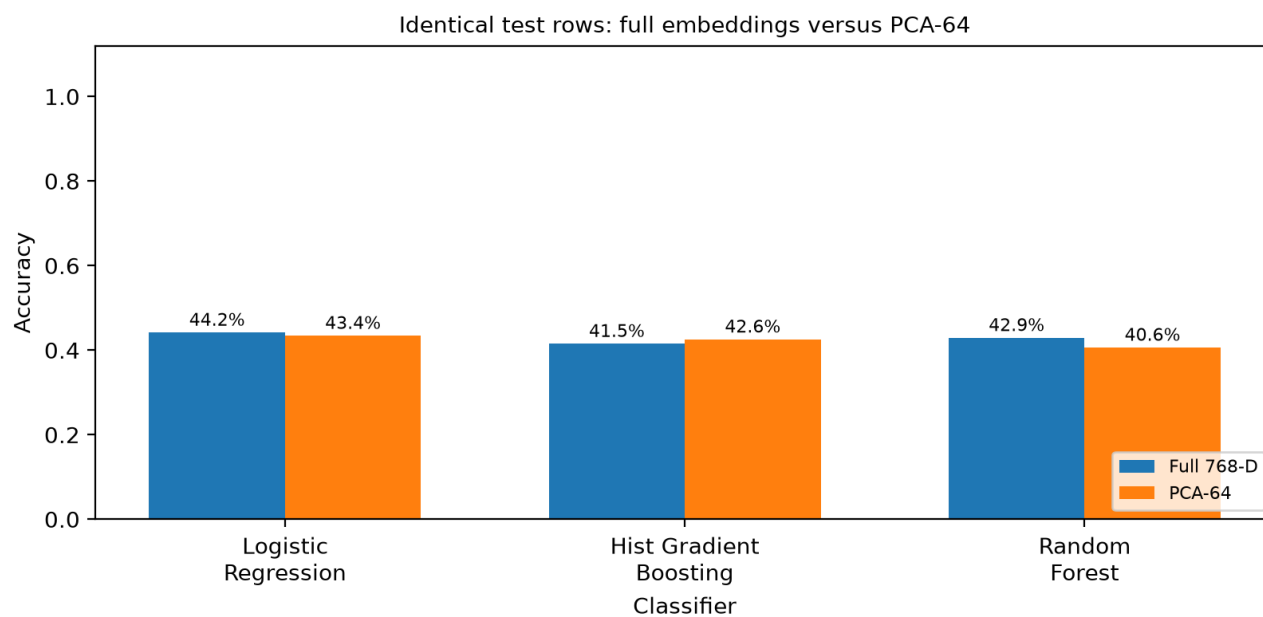
Text size	Classifier	Accuracy	Precision*	Recall*	F1*
768-D	Logistic Regression	82.29%	82.89%	83.10%	82.92%
768-D	Hist Gradient Boosting	83.64%	84.72%	84.03%	84.21%
768-D	Random Forest	77.44%	77.87%	78.24%	77.93%
PCA-64	Logistic Regression	82.44%	83.01%	83.22%	83.03%
PCA-64	Hist Gradient Boosting	82.09%	83.28%	82.54%	82.76%
PCA-64	Random Forest	77.89%	78.47%	78.65%	78.35%

*Precision, recall and F1 are macro-averaged. Fusion adds 22 patient features: $768 + 22 = 790$, or $64 + 22 = 86$ total inputs. PCA retains 91.31% of development embedding variance.

Interpretation

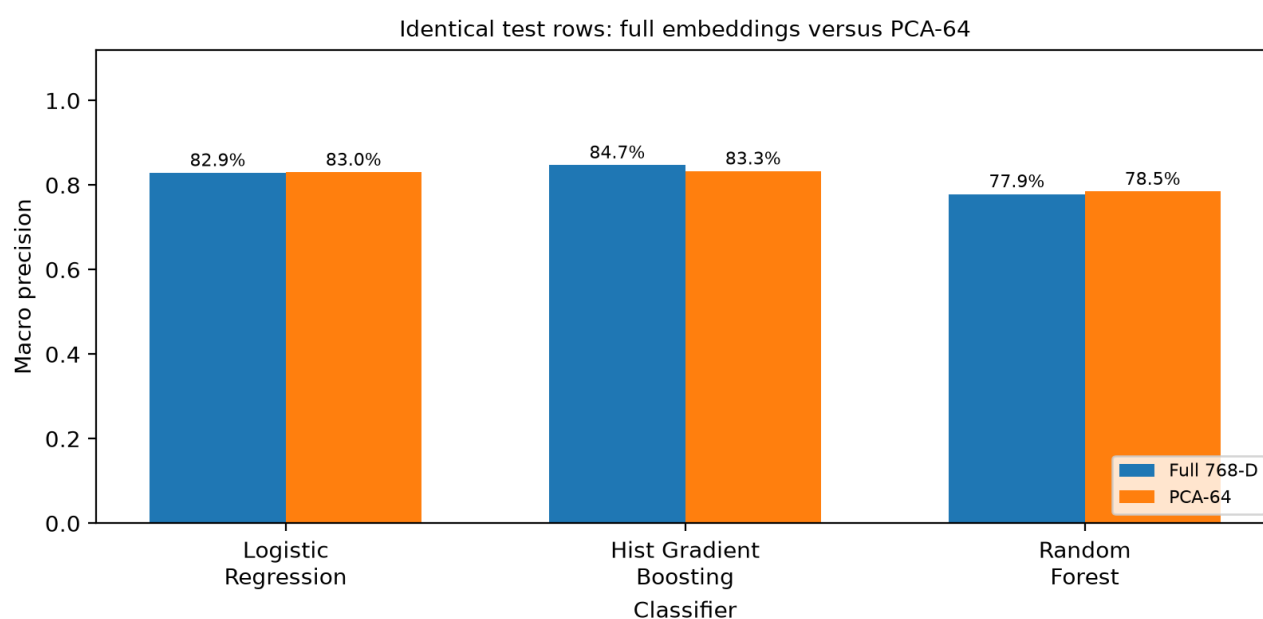
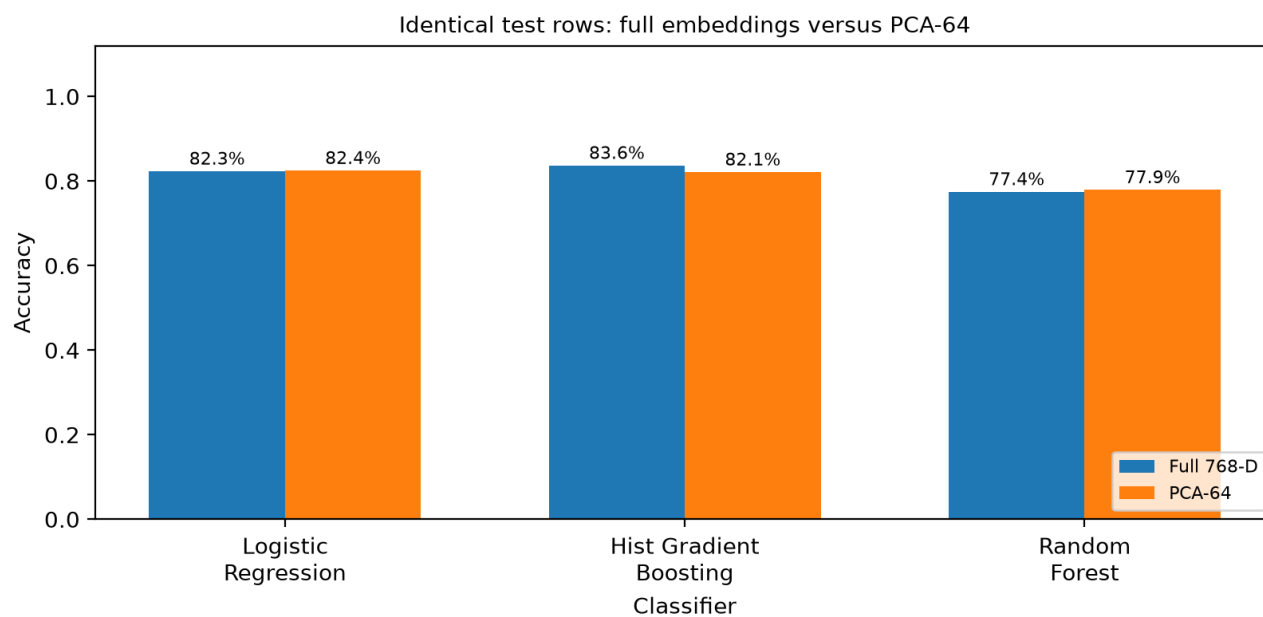
All classifiers are newly fitted to the four-level targets. The encoder is frozen, not fine-tuned. Five-fold development CV chooses the GUI model before holdout evaluation (page 8). Scores measure agreement with dataset labels, not clinical validation.

Text-only performance



Blue bars: original 768-dimensional embeddings. Orange bars: the same embeddings reduced to 64 dimensions. Scores are from the fixed test split, not mixed cross-validation results.

Performance with patient features



Blue bars: original 768-dimensional embeddings. Orange bars: the same embeddings reduced to 64 dimensions. Scores are from the fixed test split, not mixed cross-validation results.

Logistic Regression: confusion matrices

Rows show reference triage levels; columns show predictions. Every matrix contains the same 1,999 test records. Top row: text only. Bottom row: text plus patient features. Left: full 768-D. Right: PCA-64.

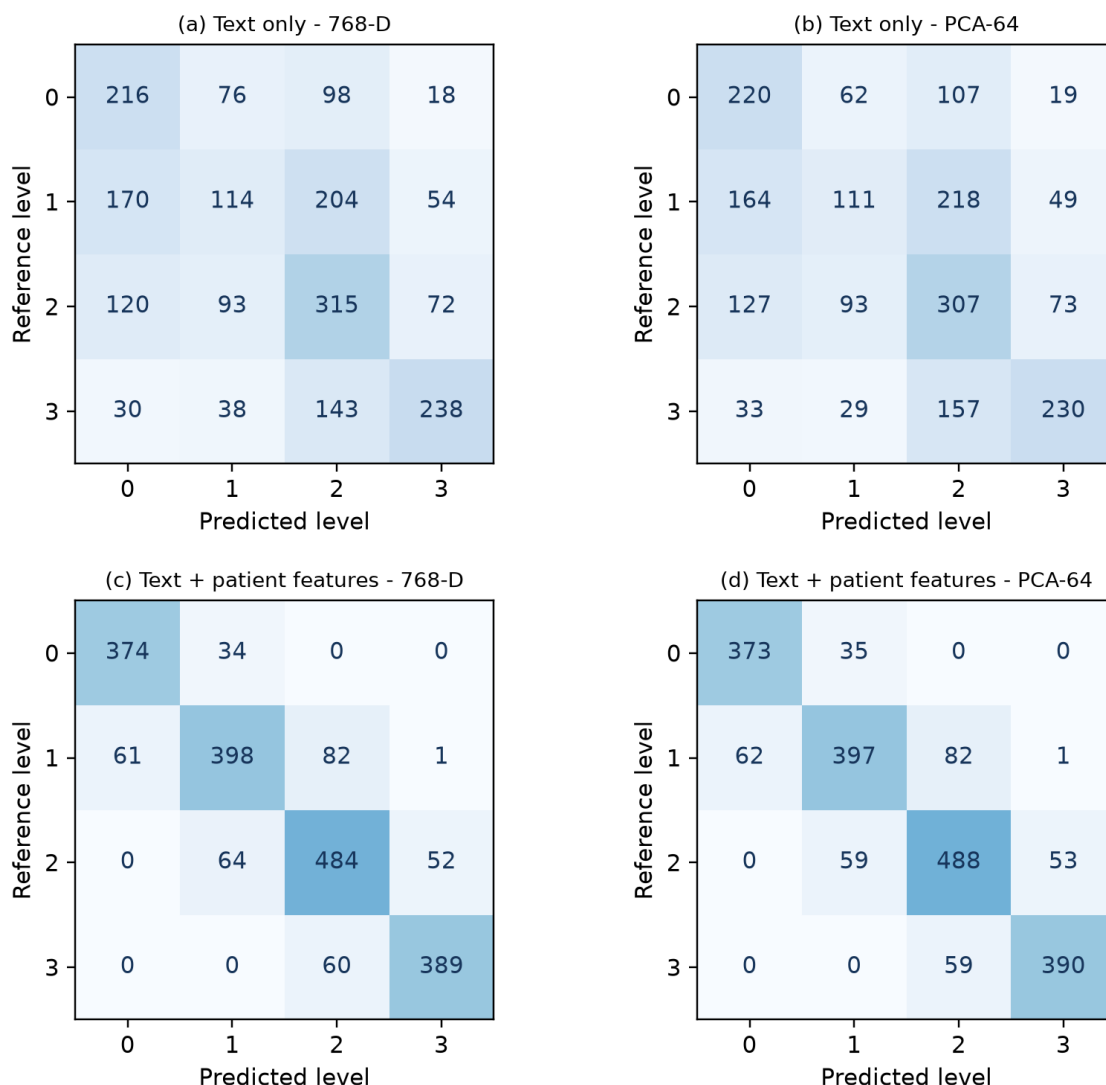


Figure: Logistic Regression under the four input conditions. Only the representation/input condition changes; the classifier settings and split stay fixed. Each matrix contains all four levels: 0 Emergency, 1 Urgent, 2 Standard and 3 Non-urgent.

Hist Gradient Boosting: confusion matrices

Rows show reference triage levels; columns show predictions. Every matrix contains the same 1,999 test records. Top row: text only. Bottom row: text plus patient features. Left: full 768-D. Right: PCA-64.

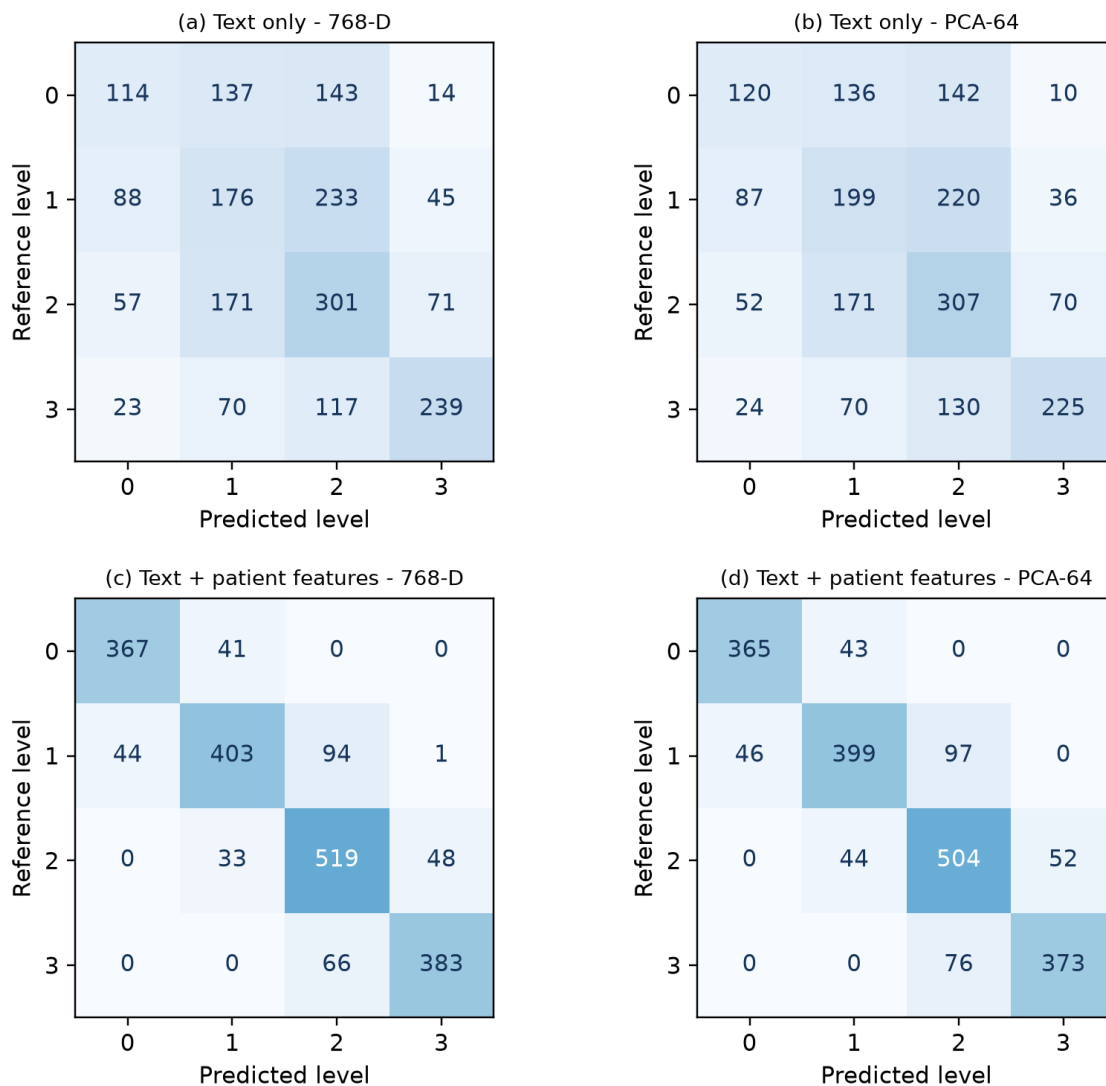


Figure: Hist Gradient Boosting under the four input conditions. Only the representation/input condition changes; the classifier settings and split stay fixed. Each matrix contains all four levels: 0 Emergency, 1 Urgent, 2 Standard and 3 Non-urgent.

Random Forest: confusion matrices

Rows show reference triage levels; columns show predictions. Every matrix contains the same 1,999 test records. Top row: text only. Bottom row: text plus patient features. Left: full 768-D. Right: PCA-64.

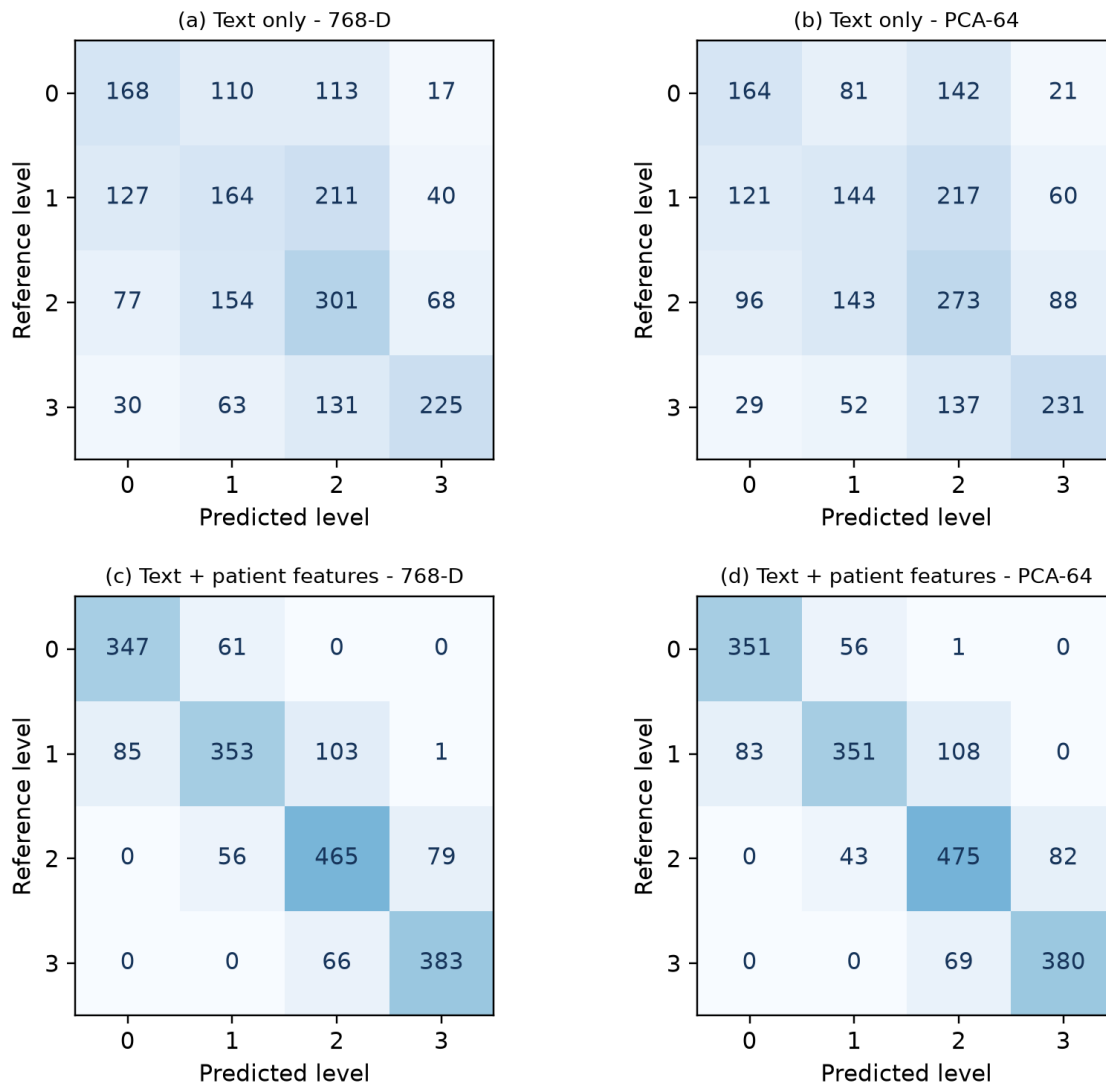


Figure: Random Forest under the four input conditions. Only the representation/input condition changes; the classifier settings and split stay fixed. Each matrix contains all four levels: 0 Emergency, 1 Urgent, 2 Standard and 3 Non-urgent.

Section 3.6 | Comparison and reproducibility

Performance comparison of ED triage models

Model	Accuracy	Recall	F1
TF-IDF + Logistic Regression [1]	0.750	0.988	0.544
TF-IDF + Random Forest [1]	0.751	0.964	0.582
BiLSTM [1]	0.746	0.846	0.670
CNN [1]	0.723	0.787	0.667
Current Logistic Regression + PCA-64	0.8244	0.8322	0.8303
Current Hist Gradient Boosting + PCA-64	0.8209	0.8254	0.8276
Current Random Forest + PCA-64	0.7789	0.7865	0.7835
Selected Logistic Regression + PCA-64	0.8519	0.8590	0.8580

Table layout follows the supplied screenshot. Current rows use combined inputs and macro recall/F1. Literature rows use a different binary KTAS task and the published metric definitions, so this is context, not a like-for-like superiority benchmark.

Fixed classifier settings

Logistic Regression: $C = 1$, balanced training weights, 2,000 maximum iterations. Hist Gradient Boosting: 300 iterations, learning rate 0.1, other settings at the uploaded baseline defaults. Random Forest: 300 trees, minimum leaf size 3, `max_features = 0.7`, balanced training weights. Random seed is 42. These settings define the 12 baseline comparisons. A separate, prespecified 10-configuration search selects the serving model on development CV only; no post-test tuning or probability adjustment is used.

Data and representation controls

Groups connect repeated normalized complaints or clinical concepts. Numeric imputation/scaling and categorical encoding are fitted on development rows. The full SVD PCA solver is fitted once per input view using development embeddings. The same encoder, labels, rows and classifier settings are retained across the four conditions.

The workbook contains 10,000 records. Its 4,290 missing concepts were recovered only after all 11 complaint/patient inputs and populated concepts matched the previous supplied file row-for-row; new labels were preserved. Triage_Label, Category and processing metadata are excluded. Source checksums, splits and all predictions are saved locally.

[1] Seo et al. (2025). Artificial intelligence for severity triage based on conversations in an emergency department in Korea. Scientific Reports 15, 16870. DOI: 10.1038/s41598-025-99874-0, Table 2. The publication uses binary KTAS and ten-fold cross-validation.

The provider described labeling as “by using all”; the exact process and independent per-record review are not documented. Evaluation uses supplied clinical concepts; live Ollama translation performance has not been established. The GUI retains the anatomical gate. A missing complaint displays 50% as a placeholder and no triage level.

Selected model | Validation and class performance

Five-fold grouped development cross-validation

Candidate	Accuracy	Macro F1 +/- SD	L0 recall	Under-triage
logreg_c10	85.46%	85.98 +/- 0.53	93.38%	7.07%
hgb_slow	83.79%	84.30 +/- 0.94	88.97%	8.96%
hgb_regularized	83.48%	83.98 +/- 0.82	90.20%	8.91%
hgb_baseline	83.18%	83.71 +/- 0.76	89.83%	9.11%
hgb_balanced	83.03%	83.55 +/- 0.83	89.83%	8.96%
logreg_baseline	82.51%	83.10 +/- 0.59	93.08%	8.39%
rf_balanced	78.83%	79.28 +/- 0.89	89.71%	11.34%
rf_baseline	78.50%	79.00 +/- 1.03	88.36%	11.71%
rf_fine	78.74%	78.92 +/- 0.34	93.93%	10.46%
logreg_regularized	77.82%	78.48 +/- 0.57	91.30%	10.40%

Selection: highest mean macro F1, with emergency recall and lower under-triage as tie breakers. PCA and patient preprocessing are fitted separately inside every training fold. SD is across folds; it is not a confidence interval.

GUI model: Logistic Regression + PCA-64

Selected settings: C=10, balance=True. Classifier seed: 42.

Held-out accuracy 85.19%; macro precision 85.71%; macro recall 85.90%; macro F1 85.80%. Under-triage 7.60%; over-triage 7.20%. Selected candidate: logreg_c10.

Group-bootstrap 95% intervals: accuracy 83.73% to 86.75%; macro F1 84.36% to 87.24% (1,000 resamples of held-out groups).

Level	Precision	Recall	F1	Support
0 Emergency	90.17%	92.16%	91.15%	408
1 Urgent	82.92%	79.70%	81.28%	542
2 Standard	81.19%	82.00%	81.59%	600
3 Non-urgent	88.57%	89.76%	89.16%	449

